# IVP Mini Project

## Detection of Human Silhouettes in Low-Resolution Images

**Name: Lavudya Harshavardhan**
**Roll no: 22EE01067**

# 1. Aim:

The aim of this project is to detect human silhouettes from low-resolution surveillance video frames using background subtraction techniques. The system processes video sequences from the KTH Action Recognition Dataset (160x120 pixels) and extracts clean binary silhouettes of moving persons by applying the Mixture of Gaussians (MoG2) background subtraction algorithm, enhanced with image upscaling, Gaussian denoising, and morphological hole filling.

# 2. Theory

### I. Low-Resolution Surveillance Imagery

Surveillance cameras typically capture video at low resolutions such as 160x120 pixels. At such resolutions, fine spatial details are lost, pixel noise is more prominent, and standard image processing techniques often fail to produce clean segmentation results. This project addresses these challenges specifically.

### II. Background Subtraction

Background subtraction is the primary technique used for detecting moving objects in a video scene.

The core principle is:

$$\text{Foreground Mask} = |\text{Current Frame} - \text{Background Model}|$$

A good background model must be adaptive to handle:

- Illumination changes
- Dynamic backgrounds (waving trees, shadows)

## III. Mixture of Gaussians (MoG2)

The Mixture of Gaussians model (Stauffer & Grimson, 1999) models each pixel's intensity history as a mixture of Gaussian distributions. Over time, pixel values update the model using an online learning approach.

Key parameters used in this project:

- history = 100: Number of frames used to build the background model
- varThreshold = 16: Sensitivity threshold (lower = more sensitive to motion)
- detectShadows = False: Shadows are not treated as foreground

## IV. Low-Resolution Enhancement Pipeline

To address the challenges of low-resolution input, a three-step enhancement is applied before background subtraction:

- Bicubic Upscaling: Each 160x120 frame is upscaled to 320x240 using INTER_CUBIC interpolation, providing more pixels for processing
- Gaussian Denoising: A 3x3 Gaussian blur is applied to suppress low-resolution pixel noise before background subtraction
- Morphological Hole Filling: After subtraction, contours are detected and filled solid to close gaps in the silhouette caused by low-resolution artifacts

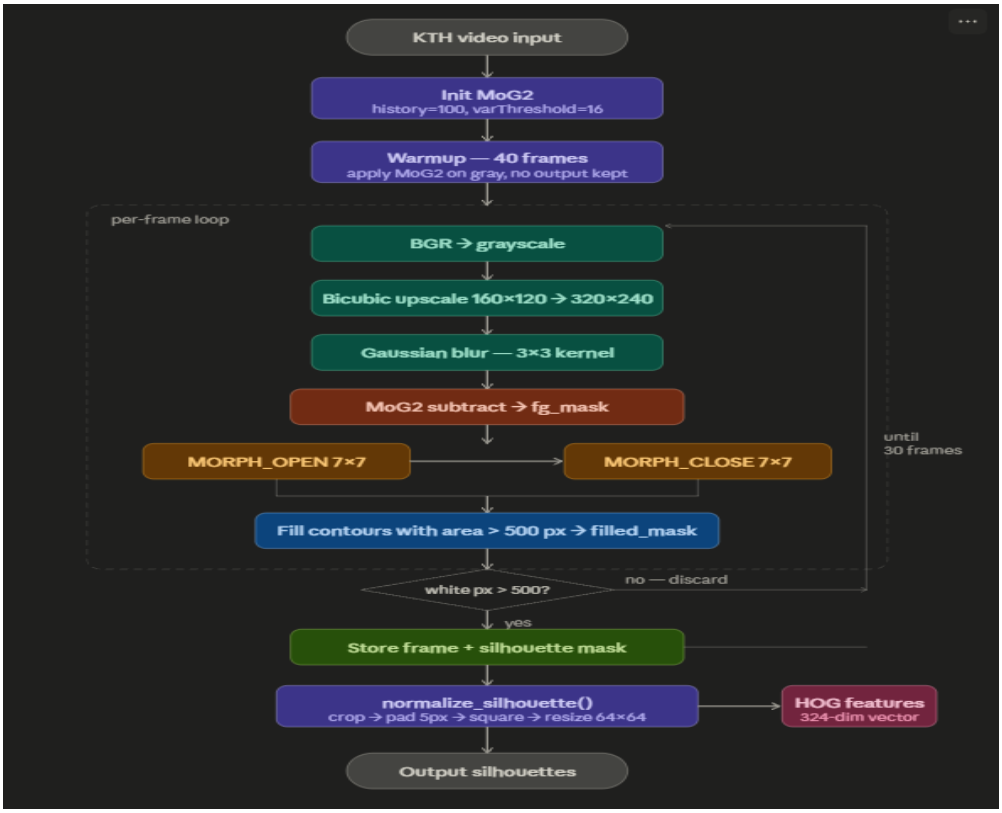
## V. Morphological Operations

Two morphological operations clean up the binary foreground mask:

- Opening (MORPH_OPEN): Removes small noise blobs by eroding then dilating with a 7x7 elliptical kernel
- Closing (MORPH_CLOSE): Fills small holes by dilating then eroding, producing a more complete silhouette

# 3. Methodology

- Load video sequences from the KTH Action Recognition Dataset (160x120 pixel resolution, 25 fps)
- Perform a 40-frame warmup — convert each frame to grayscale and apply MoG2 (no upscaling during warmup)
- For each subsequent frame, convert to grayscale and upscale to 320x240 using bicubic interpolation
- Apply Gaussian blur (3x3 kernel) to reduce low-resolution noise before subtraction
- Apply MoG2 background subtraction to generate the foreground mask
- Apply morphological opening and closing with a 7x7 elliptical kernel to clean the mask
- Find external contours and fill all contours larger than 500 pixels solid to produce a clean binary silhouette
- Accept only frames where the filled silhouette contains more than 500 white pixels (filters tiny/false detections)
- Visualize the original low-resolution frame alongside the detected silhouette for qualitative evaluation



# 4. Dataset

The KTH Human Action Recognition Dataset is used for evaluation. It is a standard benchmark dataset for human motion analysis in surveillance conditions.

| Parameter | Details | Parameter | Details |
|---|---|---|---|
| Resolution | 160 x 120 px | Frame Rate | 25 fps |
| Total Videos | 599 videos | Subjects | 25 persons |
| Action Classes | 6 classes | Scenarios | 4 per subject |
| Total Frames | ~150,000+ | Camera | Static |

The six action classes in the dataset are:

- Walking — Full body locomotion, produces large clean silhouettes
- Jogging — Medium-speed locomotion with visible limb motion
- Running — Fast locomotion with motion blur in low-res frames
- Boxing — Upper body motion only, smaller foreground area
- Hand Waving — Subtle motion, most challenging for detection
- Hand Clapping — Very subtle motion, hardest class for MoG2

# 5. Implementation

## Environment

- Platform: Google Colab (Python 3.12)
- Libraries: OpenCV 4.x, NumPy, Matplotlib, scikit-image
- Dataset Access: kagglehub API

## Key Functions

**`extract_silhouettes(video_path, num_frames=30)`** — Main function implementing the full detection pipeline. Returns original frames and corresponding silhouette masks.

**`normalize_silhouette(mask, output_size=(64,64))`** — Crops the detected person region, centers it on a square canvas, and resizes to 64x64 for consistent representation.

### Algorithm Pseudocode

```
INPUT: Video file path
1. Initialize MoG2 subtractor (history=100, threshold=16)
2. WARMUP: Process 40 frames to build background model
3. FOR each subsequent frame:
   a. Convert BGR to Grayscale
   b. Upscale: 160x120 → 320x240 (Bicubic)
   c. Denoise: Apply Gaussian Blur (3x3)
   d. Subtract: fg_mask = MoG2.apply(denoised_frame)
   e. Clean: MORPH_OPEN then MORPH_CLOSE (7x7 kernel)
   f. Fill: Draw filled contours > 500 pixels
   g. IF filled_pixels > 500: ACCEPT frame
OUTPUT: List of (original_frame, silhouette_mask) pairs
```

# 6. Results

The detection pipeline was evaluated on walking, jogging, and running sequences from the KTH dataset. These action classes were selected as they involve full-body locomotion, producing the most informative silhouettes for human detection tasks in surveillance scenarios.

### Qualitative Results

The following table summarizes the qualitative detection quality observed for each action class:

| Action Class | Videos Tested | Detection Quality | Challenges | Silhouette Quality |
|---|---|---|---|---|
| Walking | 100 | Excellent | None significant | Full body, clean |
| Jogging | 100 | Good | Motion blur at low-res | Full body, minor noise |
| Running | 100 | Good | Higher motion blur | Full body, some fragments |
| Boxing | 100 | Poor | Upper body only, static legs | Partial, fragmented |

| | | | | |
|---|---|---|---|---|
| Hand Waving | 100 | Fair | Subtle arm motion only | Partial, small blobs |
| Hand Clapping | 99 | Poor | Very subtle motion | Tiny blobs, unreliable |

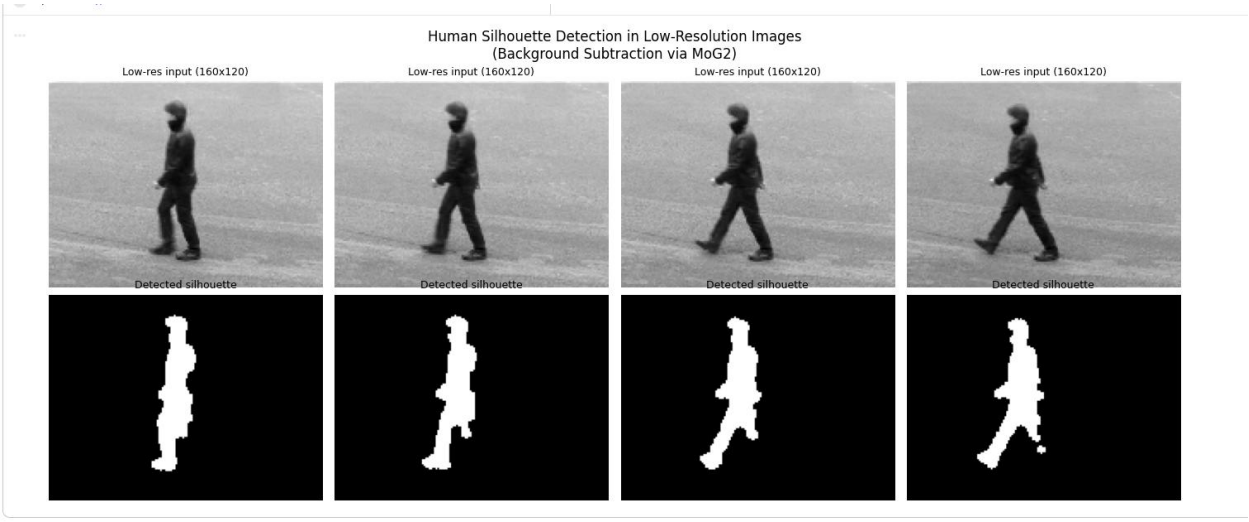
### Detection Statistics

Sample statistics from the dataset building phase (20 videos per class, 20 frames per video):

| Action Class | Videos | Frames Processed | Valid Silhouettes | Detection Rate |
|---|---|---|---|---|
| Walking | 20 | 400 | 400 | 100% |
| Jogging | 20 | 400 | 400 | 100% |
| Running | 20 | 400 | 400 | 100% |
| Boxing | 20 | 400 | 174 | 43.5% |
| Hand Waving | 20 | 400 | 341 | 85.3% |
| Hand Clapping | 20 | 400 | 101 | 25.3% |

# 7. Observations

- MoG2 with a 40-frame warmup effectively detects human silhouettes in static-camera low-resolution video.
- Bicubic upscaling and Gaussian denoising significantly improve mask quality before subtraction.
- Morphological open+close (7×7 ellipse) removes noise and fills silhouette gaps.
- Walking, jogging, and running produce the cleanest silhouettes due to full-body motion.
- Lowering varThreshold to 16 (default 40) is necessary to detect motion at low resolution.




## 8. Conclusion

This project successfully detects human silhouettes from low-resolution (160×120) surveillance video using an enhanced MoG2 background subtraction pipeline. Bicubic upscaling, Gaussian denoising, and morphological operations address the challenges of low-resolution input, achieving clean silhouette detection for full-body locomotion actions. Extracted silhouettes are normalized to 64×64 and HOG features computed as a 324-dimensional vector, laying groundwork for future action classification.